

Chapter 14: Description of Motion

Physics Summary

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Relative Motion

- A stationary object keeps in the same position. A moving object changes place or location.
- The same object, at the same time, can be at rest from a certain point of view, while being in motion from another point view. E.g., the board in your classroom is at rest relative to your classroom/classroom walls while it is movable relative to the sun.
- Therefore, motion is said to be relative.

Space Frame of Reference

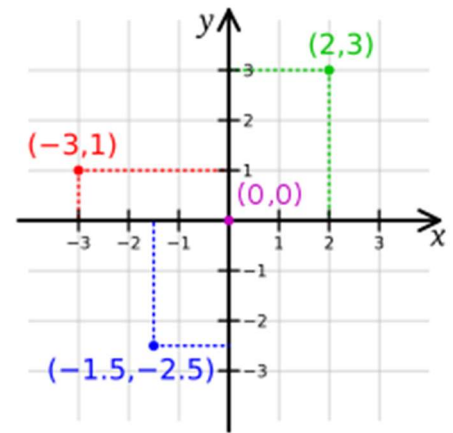
- In the above example, the classroom and the sun are called frames of reference.
- A frame of reference is an object used for comparison to determine whether other objects are moving or not.
- A frame of reference is considered as non-moving.
- A default frame of reference is the earth.

Trajectory

- The line or path followed by an object during its motion is called trajectory.
- If the mobile's trajectory is a straight line the motion is, then rectilinear.
- However, if the trajectory is curved, the motion is curvilinear.

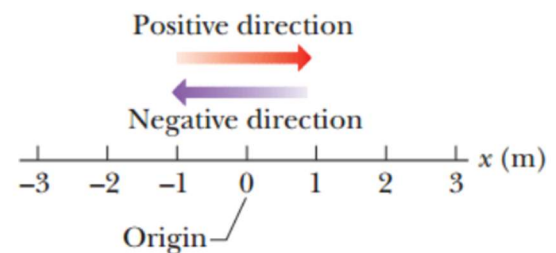
Coordinates System

- A coordinates-system is used to determine the position or location of a moving object within a certain frame of reference.
- Usually, you use in mathematics as set of two perpendicular axes, x-axis and y-axis, to tell the location of a point by its coordinates: its abscissa "x" and its ordinate "y".
- When the motion is rectilinear (one dimension), one axis is enough to tell the mobile's position. For e.g., you can consider the x-axis alone to tell the location of a train moving horizontally on straight rails. In case of an elevator, a y-axis is enough to localize the elevator during its vertical upward and downward movement.



Origin of Space

- On the x-axis, the point which is characterized by $x = 0$ is called the origin of space (O).
- The starting point is normally taken as origin of space.



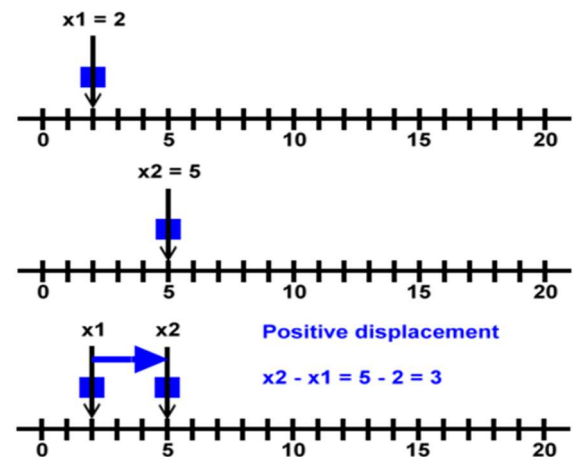
Position

- The position of the mobile M is determined by its abscissa x_M , where: $x_M = \overline{OM}$.
- The SI unit of x is meters (m).

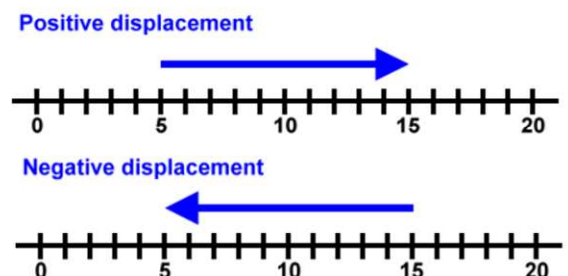
Displacement

- Displacement is the change in position of the mobile:
 $\text{Displacement} = \text{final position} - \text{initial position}$
or: $\Delta x = x_f - x_i$

- Δx is also like x , measured in (m).



- Δx can be $\begin{cases} > 0 \text{ if } x_f > x_i \text{ (motion occurs to the right)} \\ < 0 \text{ if } x_f < x_i \text{ (motion occurs to the left)} \\ = 0 \text{ if } x_f = x_i \text{ (object returns to same location)} \end{cases}$



Time Frame of Reference

- It is used to tell the date of a certain event.
- The watch on the wall of your room can be used as a time reference.



Duration

- The time interval between two instants is called duration: $\Delta t = t_f - t_i$.
- Like t , Δt is counted in SI in seconds (s).

Origin of Time

- Normally, the starting or initial time is taken as origin of time (i.e., $t_i = 0$).

Speed

- Speed tells how fast an object is.
- Speed is distance traveled per unit of time.



Average Speed

- Average speed tells the overall speed of an object during a trip.
- It is like a student's average on a report card.
- $Average\ speed = \frac{distance\ traveled}{total\ time\ taken}$ or $v_{av} = \frac{d}{\Delta t}$
- d (m) and Δt (s) so v in (m/s).

Instantaneous Speed

- The speed of a mobile at particular instant is called its instantaneous speed.
- As you look at a car speedometer, you can tell then its instantaneous speed.
- Instantaneous speed is approximately equal to average speed which is calculated over a narrow time interval:

$$v_{inst} \approx v_{av} \text{ (with } \Delta t \ll \text{)}$$

- In chronophotography, to determine the instantaneous speed at the point M_i , we use the formula:

$$v_i = \frac{M_{i-1}M_{i+1}}{t_{i+1} - t_{i-1}} = \frac{M_{i-1}M_{i+1}}{2\tau}$$

Where:

- M_{i-1} and M_{i+1} are the positions just before and after M_i .
- t_{i-1} and t_{i+1} are respectively their corresponding dates.
- " τ " is the constant time interval separating any two successive points.

Velocity

- Velocity is speed in a given direction.
- Velocity gives us both information: how fast an object moves and in which direction it does.

Representing Instantaneous Velocity Vector, $\vec{v}$

- Any vector has four elements or characteristics: origin or point of application (point O, point A ...), line of action (horizontal, vertical, oblique, ...), direction (to the right, to the left, upward, downward, ...), magnitude or length.
- The instantaneous velocity vector of a mobile is characterized by:
 - Its origin (or point of application): the current position M of the mobile (the mobile's position at the considered instant).
 - Line of action: the axis of motion itself.
 - Direction: same as the direction of motion.
 - Magnitude: the value of the instantaneous speed at the considered moment.

Acceleration

- Acceleration refers to a change in velocity with respect to time.
- Acceleration is the rate at which velocity changes: it is the change of velocity per unit of time.
- In this grade, moving objects follow rectilinear paths, as a result the motion direction cannot change only speed can do.

Average Acceleration

- *Average acceleration* = $\frac{\text{change in speed}}{\text{time taken or duration}}$ or $a_{av} = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{t_f - t_i}$
- v (m/s) so Δv (m/s); and Δt (s), then a in (m/s/s or m/s²).

Instantaneous Acceleration

- The acceleration of a mobile at a specific moment is called instantaneous acceleration.
- Instantaneous acceleration is approximately equal to average acceleration which is calculated over a narrow time interval:

$$a_{inst} \approx a_{av} \text{ (with } \Delta t \ll \text{)}$$

- In chronophotography, to determine the instantaneous acceleration at the point M_i , we use the formula:

$$a_i = \frac{v_{i+1} - v_{i-1}}{t_{i+1} - t_{i-1}} = \frac{v_{i+1} - v_{i-1}}{2\tau}$$

Where: v_{i-1} and v_{i+1} are the speeds at the points M_{i-1} and M_{i+1} surrounding M_i .

Representing Acceleration Vector, $\vec{a}$

- The instantaneous acceleration vector of a mobile is characterized by:
 - Its origin (or point of application): the current position M of the mobile (the mobile's position at the considered instant).
 - Line of action: the axis of motion itself.
 - Direction: can be the same as the velocity vector or opposite to its direction.
 - Magnitude: the value of the instantaneous acceleration at the considered moment.

Nature of motion

There are 2 cases. When $\vec{a}$ and $\vec{v}$ have:

- same direction, there is speeding up and the motion said to be accelerated.
- opposite directions, there is slowing down (decelerated motion).